

Opposition Agents Playing Magic: The Gathering

Knowledge-Graph-Grounded JEPA World Models and Active-Inference LLM Agents

Felix Neubürger¹ and Jonas Neubürger¹

¹Independent Research , github.com/cuteredpwnda/opposition-agents-playing-mtg

April 2026

Abstract

Magic: The Gathering (MTG) is, by several measures, the most combinatorially complex commercial card game ever published: more than 28,000 unique cards, a 260-page rule book, partial observability over 50–90 hidden cards, an unbounded action space, a turn-based stack with priority, and a state space conservatively estimated above 10^{800} . We present OPPOSITION AGENTS PLAYING MTG, an open-source research framework that combines (i) a from-scratch implementation of the Comprehensive Rules game engine, (ii) a Neo4j property-graph database holding a knowledge graph (KG) backed by an OWL 2 (Web Ontology Language, version 2) ontology of cards, mechanics, archetypes and combos, (iii) a four-component world model in the Vision–Memory–Controller (V+M+C) tradition of [Ha & Schmidhuber \(2018\)](#) extended with a Joint-Embedding Predictive Architecture (JEPA) head in the spirit of [LeCun \(2022\)](#), and (iv) Active-Inference Large-Language-Model (LLM) agents that fuse symbolic graph queries, latent imagination and free-energy-minimising action selection. The framework treats one seat of an MTG game as a Partially Observable Markov Decision Process (POMDP) embedded inside an n -player Markov game, and we derive each agent as an explicit approximation to that POMDP. We describe the system end-to-end, justify the design choices against the structure of the game, and report on a champion-vs-challenger self-play promotion loop with Elo (Elo, 1978) rating tracking. The code, ontology, and this report are released under the Massachusetts Institute of Technology (MIT) licence.

Keywords: world models, JEPA, knowledge graphs, active inference, LLM agents, self-play, neuro-symbolic reasoning, Magic: The Gathering.

1 Introduction

Reinforcement-learning (RL) research on games has historically progressed through a sequence of increasingly hard environments: backgammon ([Tesauro, 1995](#)), chess and Go ([Silver et al., 2017, 2018](#)), StarCraft II ([Vinyals et al., 2019](#)), and no-limit Texas hold’em poker ([Brown & Sandholm, 2018](#)). Each step pushed the community to invent new tools — temporal-difference (TD) learning, deep value networks, self-play, counterfactual regret minimisation (CFR). *Magic: The Gathering* (MTG) sits beyond the union of these challenges (Table 1): its action space is unbounded, its rules are open-textured natural language printed on cards (the comprehensive rule book itself defers to the per-card text in many places), its hidden information includes the entire opponent’s deck composition, and its mechanics are extended every quarter by the publisher Wizards of the Coast. A single MTG card can override the rules of the game (e.g. *Yixlid Jailer*, *Blood Moon*, *Cumulative Upkeep*), and a single decision in combat may resolve dozens of triggered abilities in a player-chosen order. The state space is open in the strict sense:

Table 1: Approximate combinatorial complexity comparison.

	Chess	Go	No-limit Poker (HU)	MTG
State space	10^{47}	10^{170}	10^{160}	$> 10^{800}$
Branching factor	~ 35	~ 250	~ 10	0–100+ (unbounded)
Hidden information	none	none	2 cards	50–90+ cards
Card / piece pool	6 types	1 type	52 cards	$> 28,000$ unique
Rule text length	$\sim 10\text{p}$	$\sim 5\text{p}$	$\sim 10\text{p}$	$\sim 260\text{p}$ + per-card
Rules update frequency	decades	decades	decades	quarterly

every quarterly set adds previously-unseen card text whose semantics must be parsed at runtime, so any agent that hard-codes its action vocabulary cannot survive a set rotation.

For a concrete sense of why this matters for an algorithmic agent: in a single midgame turn the active player may need to (i) decide whether to play a land, (ii) sequence two or three spells with mana cost, colour requirements and stack ordering, (iii) optionally activate any number of permanent abilities that themselves go on the stack, (iv) declare attackers under summoning-sickness and tapping rules, (v) respond to triggered abilities resolving in the order chosen by their controllers, and (vi) decide which damage to assign in combat under multi-blocker rules. Each of these is itself a sub-decision tree. The total number of legal actions per priority pass routinely exceeds 60 and, in pathological combo states, several hundred. A model-free policy that emits a fixed-size logit vector cannot represent this branching, and a tabular search procedure that enumerates the subtree cannot fit it into memory.

A purely model-free deep-RL approach is therefore unattractive: the action space cannot be enumerated as a fixed-size logit vector, the observation space is symbolic and irregular, and every new set introduces previously-unseen card text whose semantics must be parsed at run time. Conversely, a purely symbolic LLM-prompting approach struggles with rules-correctness, hallucinated cards, and the long-horizon credit-assignment that competitive play requires — an LLM that simply names a move it would like to make has no guarantee that move is legal, that it resolves the way the LLM expects, or that it furthers a coherent multi-turn plan.

Our position is that the structure of MTG demands a *neuro-symbolic* stack: a verified rules engine to keep play legal, a knowledge graph to make card semantics queryable, a learned world model to enable imagination and planning over the resulting compressed state, and an LLM agent that orchestrates these components under an active-inference objective. Figure 1 sketches the resulting architecture as a data-flow diagram, separating the *symbolic* substrate (data sources, engine, knowledge graph) from the *learned* substrate (world model, agent zoo) and showing where information actually crosses the boundary.

Contributions. This work makes four contributions.

1. A scientific framing of MTG as an *open-ended partially observable control problem* whose difficulty comes not only from combinatorics, but from the continual introduction of new symbolic rules through quarterly card releases.
2. A neuro-symbolic research substrate that couples a rules-correct engine, an ontology-grounded knowledge graph, and a latent world model, making it possible to study planning under both symbolic and statistical uncertainty in one environment.
3. An *active-inference reading* of MTG decision making in which pragmatic value, epistemic value, and language-model priors can be combined inside a single action-selection objective rather than treated as unrelated heuristics.

entire agent population.

Scope and what we do *not* do. We do not solve MTG; we do not even target tournament-strength play. The contribution is a research substrate that is rules-correct on the implemented card pool, that maintains a queryable symbolic representation of card semantics, and that supports plug-in research on world models and inference-style agents on a problem qualitatively harder than the standard benchmarks. The empirical claims in §6 are therefore framework-level (decks complete, agents converge faster than baselines, ablations show graded contributions) rather than absolute strength claims against human play.

The framework is released as open source¹ and the codebase is the canonical artifact accompanying this report. Section 2 positions the work against prior literature; §3 formalises the POMDP and the four learning principles (world models, FEP, knowledge-graph priors, constrained-decoding LLMs); §4 walks the architecture top-down; §6 presents the empirical protocol and preliminary results; §7–8 discuss limitations and next steps.

2 Related Work

World models. Ha & Schmidhuber (2018) introduced the V+M+C decomposition — a vision-VAE, a recurrent dynamics model with mixture-density output, and a tiny linear/MLP controller trained in the dream of M. Successors include Dreamer (Hafner et al., 2020, 2023) and the Schmidhuber-line of self-referential systems (Schmidhuber, 2015; Irie et al., 2022). We adopt the V+M+C skeleton but (i) replace the convolutional vision encoder with a Set-Transformer over symbolic game tokens, and (ii) attach a JEPA head (LeCun, 2022; Assran et al., 2023) that predicts a target encoding rather than a pixel-level reconstruction — a closer fit to MTG, where “imagination” should preserve abstract game state, not card art. On the software side we now follow the modular research stack exemplified by `stable-pretraining`, `stable-worldmodel`, and `LeWorldModel` (Balestriero et al., 2025; Maes et al., 2026a,b): the MTG-specific encoder, dynamics, and controller remain local to this repository, while stage-5 JEPA optimisation is delegated to the upstream training infrastructure.

Active inference. The Free-Energy Principle of Friston (2010) and its discrete-state operationalisation (Sajid et al., 2021) provide a principled objective for action selection under uncertainty: minimise expected free energy, which decomposes into a pragmatic (reward-seeking) and an epistemic (information-seeking) term. We use this decomposition to balance “play to win” against “probe the opponent” decisions during the early game.

Knowledge graphs and GraphRAG. Neo4j with the `n10s` plugin enables direct ingestion of OWL ontologies as labeled property graphs (Neo4j Labs, n.d.); the APOC library adds graph algorithms; GraphSAGE (Hamilton et al., 2017) provides inductive node embeddings. GraphRAG (Edge et al., 2024) has recently emerged as a complement to vector RAG for queries that benefit from explicit relational structure. MTG is a textbook GraphRAG case: “find me a two-card combo using only cards in my library that survive Counterspell” is a multi-hop graph query, not a similarity search.

MTG and other complex card games. Earlier MTG-AI work has used hand-crafted heuristics (Cowling et al., 2012), MCTS, and small policy nets on restricted card pools (e.g., `open-mtg`). On the rules side, `Forge`, `XMage` and `Mage` provide JVM-based reference implementations. To our knowledge, no prior public system combines a full Comprehensive-Rules engine with a learned world model and a graph-grounded LLM controller. Moravčík et al. (2017) (poker),

¹<https://github.com/cuteredpwnda/opposition-agents-playing-mtg>

Vinyals et al. (2019) (StarCraft) and Schrittwieser et al. (2020) (MuZero) are spiritual antecedents at the algorithmic level.

Ontologies for games. Game ontologies have been explored for serious games and educational metadata; for trading-card games specifically, public OWL ontologies are sparse. Our ontology (`data/ontology/mtg-ontology-v1.1.owl`) is hand-crafted following the methodology surveyed in Noy & McGuinness (2001) and validated by SHACL shapes (Knublauch & Kontokostas, 2017). The framework is designed to interoperate with the OntologyExtender tool (Data Science Lab FHSWF, 2024) — a separate, in-development multi-agent human-in-the-loop ontology-evolution toolchain by the same lab — but in the present version the OWL schema is a fixed v1.1 artefact: we do not currently run automated extension passes.

3 Theoretical Foundations

3.1 MTG as a partially-observable Markov game

We model a game of Magic from the perspective of a single seat as a finite-horizon POMDP $(\mathcal{S}, \mathcal{A}, \mathcal{O}, T, \Omega, R, \gamma, b_0)$ embedded inside an n -player Markov game. The full state $s_t \in \mathcal{S}$ encompasses every public and private zone of every player — libraries, hands, the stack, mana pools, and a long tail of bookkeeping flags (summoning sickness, monarchy, the day/night state, designations, emblems, dungeon room, companion eligibility). A controlled-perspective filter $\Omega : \mathcal{S} \rightarrow \mathcal{O}$ hides the opposing players’ hands and the order of their libraries; the agent only ever observes $o_t = \Omega(s_t)$. The action space $\mathcal{A}_t = \text{LEGAL}(s_t)$ is computed afresh on every priority pass and is highly variable: typically $|\mathcal{A}_t| \in [1, 60]$, but can spike past several hundred in hard combo states. The transition model $T(s' | s, a)$ is the rules engine itself, fully deterministic given a random seed and the active player’s choices. Reward R is sparse and terminal — ± 1 on the win condition — but agents may shape it during training using Δlife , Δcards , Δtempo as auxiliary signals. The state space is combinatorially large: a conservative count over deck-times-zone configurations gives $|\mathcal{S}| > 10^{800}$, several orders of magnitude beyond Go.

The optimal value function over beliefs $b_t \in \Delta(\mathcal{S})$ satisfies the Bellman equation

$$V^*(b) = \max_{a \in \mathcal{A}} \left[\sum_s b(s) R(s, a) + \gamma \sum_o \Pr(o | b, a) V^*(b'_o) \right], \quad (1)$$

where b'_o is the Bayes update of b after taking a and observing o . Exact solution is intractable. Each agent in our zoo is therefore an explicit approximation that trades different axes of accuracy: tabular reactive policies (Random, Heuristic, KG-Heuristic, LLM) collapse b_t to a point estimate $s_t \approx o_t$; the world-model agent learns an amortised Bellman residual in latent space; the active-inference agents maintain an explicit belief and minimise expected free energy.

3.2 World models as latent imagination

The V+M+C decomposition of Ha & Schmidhuber (2018) factors an agent into a perception module V , a recurrent dynamics module M , and a controller C . We follow this skeleton but adopt LeCun and Assran’s recommendation (LeCun, 2022; Assran et al., 2023) to predict in *representation space* rather than in observation space: the model is trained to match its H -step rollout of the latent against a target encoder’s output, not to reconstruct cards or board art. Concretely the training loss for the dynamics is

$$\mathcal{L}_{\text{JEPA}} = \sum_t \|\hat{z}_{t+1} - \text{sg}(z_{t+1})\|_2^2 + \beta \cdot \text{KL}(q(z_t | o_t) \| p(z_t | h_{t-1})), \quad (2)$$

where `sg` is stop-gradient on the target branch, h_{t-1} is the previous LSTM hidden state of M , and β is the single tunable hyperparameter we expose (`-jepa-beta`, default 1.0). At decision time the controller scores each candidate action by a $K \times H$ Monte-Carlo rollout in latent space:

$$\text{score}(a) = \frac{1}{K} \sum_{k=1}^K V_{\theta}(h_{t+H}^{(k)} | a_t = a), \quad (3)$$

with $K = 8$, $H = 10$ by default. No actual rules-engine simulation is performed during inference; this is what makes the agent fast despite the engine being heavy. The controller therefore amortises the Bellman residual: C is implicitly approximating $\arg \max_a Q^*(s_t, a)$ given an imperfect $\hat{T}$.

Training implementation note. The current stage-5 training path keeps the MTG architecture in `src/world_model/world_model.py` but wraps optimisation in `stable_pretraining.Manager` and Lightning callbacks rather than a bespoke trainer. We keep `stable-worldmodel` in the stack as the upstream world-model research substrate and benchmark interface, matching the layering used by LeWorldModel (Maes et al., 2026a; Balestrieri et al., 2025; Maes et al., 2026b) while preserving MTG-specific state encoding, graph fusion, and legal-action masking locally.

3.3 Active inference and the free-energy principle

Friston’s free-energy principle (Friston, 2010) frames perception and action as two sides of variational inference under a generative model $p(o, s)$. Variational free energy decomposes as

$$F[q] = \mathbb{E}_q[\log q(s) - \log p(o, s)] = \text{KL}(q(s) \parallel p(s | o)) - \log p(o), \quad (4)$$

so minimising F over q simultaneously sharpens beliefs (the KL term) and accumulates evidence ($-\log p(o)$). Action selection is recovered by minimising *expected* free energy, conditioned on a policy $\pi = (a_t, a_{t+1}, \dots)$:

$$G(\pi) = \underbrace{-\mathbb{E}_q[\log p(o_{\tau} | C)]}_{\text{pragmatic value}} + \underbrace{-\mathbb{E}_q[\text{KL}(q(s_{\tau} | o_{\tau}, \pi) \parallel q(s_{\tau} | \pi))]}_{\text{epistemic value}}, \quad (5)$$

where C encodes the agent’s preferences (in our setting: “win”), $q(s_{\tau} | \pi)$ is the predictive belief under policy π , and the epistemic term rewards information-gathering actions. Bayesian RL falls out as the special case where preferences are uniform; pure exploration falls out when rewards vanish. In the MTG context, the pragmatic term encodes “play to win” and the epistemic term encodes “probe the opponent’s hand” — exactly the trade-off that Sajid et al. (2021) formalise. Our `ActiveInferenceAgent` maintains a Dirichlet belief over the opponent’s hand composition seeded from the decklist (cf. `docs/OPPONENT_MODELING_WITH_DECKLIST.md`) and re-evaluates Eq. (5) at every priority pass.

3.4 Knowledge graphs as a symbolic prior

Cards form an inductive graph $\mathcal{G} = (V, E, \tau)$ with typed edges $\tau \in \{\text{combo}, \text{synergy}, \text{counter}, \text{enables}\}$ ingested from the OWL ontology and an embedding pipeline trained with GraphSAGE (Hamilton et al., 2017). GraphSAGE is inductive: a freshly printed card v_{new} obtains an embedding $\mathbf{h}_{v_{\text{new}}} \in \mathbb{R}^{128}$ by sampling and aggregating its neighbours, without retraining. We exploit this in two ways: (i) the encoder feeds a graph latent $g_t = \text{Enc}_{KG}(\mathcal{G}, s_t)$ into the recurrent dynamics so that the world model’s prior is graph-aware, and (ii) the heuristic agent rescores candidate spells by their graph distance to combo completions, mirroring GraphRAG’s “relational evidence” pattern (Edge et al., 2024).

In implementation, we separate immutable source facts (card oracle data, ontology-grounded edges) from learned extension evidence. Self-play enrichment appends provenance-linked learned nodes/edges instead of mutating base facts, allowing many agents and runs to contribute to one shared strategic memory while preserving reproducibility and auditability.

3.5 LLM as a constrained-decoding policy

The LLM never freely generates moves. Given the prompt $\rho(s_t, \mathcal{A})$ that enumerates legal actions by index, we read off

$$\log p_{\text{LLM}}(a \mid s_t) = \text{LM}(\text{idx}(a) \mid \rho(s_t, \mathcal{A})), \quad (6)$$

masking any token outside the integer range $\{0, \dots, |\mathcal{A}| - 1\}$. This recovers the standard “constrained decoding” guarantee — the LLM cannot hallucinate an illegal move — while still letting it import its general-knowledge prior. Calibration of the resulting distribution is controlled by a temperature parameter (0.2 in our default config), which interpolates between greedy LLM picks and a uniform fallback.

3.6 Self-play promotion as fixed-point iteration

Training combines all three signals. The replay buffer $\mathcal{D}$ is collected by self-play between agent variants; the world model is fit on $\mathcal{D}$; and the controller is improved iteratively by the champion-vs-challenger schema implemented in `scripts/train_pipeline.py`. At each iteration i , a challenger π_{i+1} is trained against π_i on the latest replay window and faces the champion in a best-of- N match; promotion happens iff the empirical win rate exceeds 0.55. This is a stochastic-approximation fixed-point operator on the policy improvement step (Tesauro, 1995; Silver et al., 2017, 2018; Schrittwieser et al., 2020): under standard Robbins–Monro conditions it converges, in expectation, to a fixed point of the Bellman optimality operator restricted to the dynamics class $\hat{T}$ that the world model has learned. Convergence in practice depends on $\hat{T}$ being locally accurate, which we monitor by tracking JEPA validation loss across iterations.

4 System Architecture

4.1 Game engine

The engine implements the Comprehensive Rules subset relevant for two-player Standard plus the structural extensions for multiplayer Commander. Two parallel orchestrators sit on top of the same dataclasses: a synchronous `GameSimulator` (`src/engine/game_simulator.py`) used in tournaments and tests, and an asynchronous `GameRunner` (`src/orchestrator/game_runner.py`) used in self-play data collection. Both implement the same end-of-game rules. Core modules:

- `game_state`: immutable-by-convention dataclasses for `GameState`, `PlayerState` (carrying `max_hand_size` and `mulligans_taken`), and `CardInstance`;
- `zones`, `stack`, `phases`, `priority_loop`: APNAP priority and the seven-step turn structure;
- `rules_engine`: legal-action generation given current priority and phase;
- `state_based_actions` (CR 704), `triggered_abilities` (CR 603), `replacement_effects` (CR 614), `continuous_effects` (CR 613 layer system);
- `combat`, `mana`, `keywords`, `abilities`: per-mechanic logic;
- **London mulligan** at game setup: draw 7, optionally re-mulligan up to a configurable cap, then bottom cards equal to the number of mulligans taken (CR 103.5);
- **Cleanup-phase discard** down to `PlayerState.max_hand_size` (default 7), which had been a long-standing gap in earlier engine snapshots (CR 514);

Comprehensive Rules concept	CR section	Implementing module
Turn structure / phases / steps	500–514	src/engine/phases.py
APNAP priority & priority loop	116–117	src/engine/priority_loop.py
The stack & spell resolution	405–608	src/engine/stack.py
Layer system (continuous effects)	613	src/engine/continuous_effects.py
Replacement effects	614–616	src/engine/replacement_effects.py
Triggered abilities	603	src/engine/triggered_abilities.py
State-based actions	704	src/engine/state_based_actions.py
Combat (declare, block, damage steps)	506–510	src/engine/combat.py
Mana & cost payment	106, 601–602	src/engine/mana.py
London mulligan	103.5	src/engine/game_setup.py
Cleanup discard to hand size	514	src/engine/phases.py::cleanup_step
Loss from empty library on draw	104.3c, 704.5b	src/engine/state_based_actions.py
Legal-action enumeration	— (engine-internal)	src/engine/rules_engine.py

Table 2: Mapping from Comprehensive-Rules concepts to the implementing engine module. Every cell is a real path; auditing a rule reduces to opening that file. The legal-action enumerator (`rules_engine.py`) is the only point of contact between the engine and any agent.

- **Deterministic empty-library loss:** drawing from an empty library ends the game with that player losing (CR 104.3c / 704.5b), in both the sync simulator and the async runner;
- **Deterministic timeout tie-breaker:** instead of auto-draw on hitting the configured turn limit, the leader on the lexicographic tuple (life total, permanents in play, hand size, library size) wins; only true ties are recorded as draws.

The engine is pure-Python, deterministic given an RNG seed, and has no LLM dependency, which makes it the ground truth that all agents play against.

Table 2 maps each Comprehensive-Rules concept to the exact module that implements it; readers reproducing the system can use it as a reading order through the codebase.

4.2 Knowledge graph

Cards are imported from Scryfall and reified as (`:Card`) nodes carrying mana cost, type line, oracle text, colour identity, and a SBERT embedding of the oracle text. Mechanics, abilities, archetypes and combos become first-class nodes connected by typed edges (`HAS_KEYWORD`, `ENABLES_COMBO`, `ARCHETYPE_OF`, ...). The OWL ontology (v1.1) declares the schema; `n10s` imports it; `pyshacl` validates it; `APOC` provides PageRank, community detection and shortest-path queries used in agent reasoning. ?? 1 shows a typical multi-hop query used by the combo-detector agent.

Listing 1: Two-card combo query.

```
MATCH (c1:Card)-[:ENABLES_COMBO]->(combo:Combo)<-[:ENABLES_COMBO]-(c2:Card)
WHERE c1.name IN $library AND c2.name IN $library AND c1 <> c2
RETURN combo.name, combo.win_condition, c1.name, c2.name
ORDER BY combo.cmc_total ASC LIMIT 10;
```


4.3 World model

Let $s_t \in \mathcal{S}$ be the game state at decision step t and $g \in \mathcal{G}$ the player’s KG context (deck, archetype tag, known opponent information). The world model decomposes as:

$$z_t = \text{Enc}_V(s_t) \quad (\text{state encoder, Set-Transformer + VAE}) \quad (7)$$

$$g_t = \text{Enc}_{\text{KG}}(g, s_t) \quad (\text{KG context encoder}) \quad (8)$$

$$h_t, \pi_t^M = \text{LSTM}_M([z_t; g_t], h_{t-1}) \quad (\text{MDN-LSTM dynamics}) \quad (9)$$

$$\hat{z}_{t+1} = \text{JEPA}([z_t; g_t; a_t]) \quad (\text{predictor head}) \quad (10)$$

$$a_t = C(z_t, g_t, h_t) \quad (\text{controller}). \quad (11)$$

Enc_V produces $z_t \in \mathbb{R}^{256}$. The VAE term acts as a regulariser; the JEPA head predicts the *encoder output* z_{t+1} , not a reconstruction, and is trained with $\text{MSE}(z_{t+1}, \hat{z}_{t+1}) + \beta \text{KL}(q_\phi \| p)$. Surprise at step $t + 1$ is the predictor residual $\|\hat{z}_{t+1} - z_{t+1}\|_2$, which seeds curiosity bonuses and KG-enrichment triggers.

Concrete tensor shapes. For reproducibility we record the default dimensions used in `src/world_model/world_model.py`: state tokens $x_t \in \mathbb{R}^{B \times N \times 64}$ with $N \leq 256$ (zero-padded with an attention mask), latent $z_t \in \mathbb{R}^{B \times 256}$, KG context $g_t \in \mathbb{R}^{B \times 64}$ produced by mean-pooling a two-layer GraphSAGE over the cards in the player’s library, LSTM hidden $h_t \in \mathbb{R}^{B \times 512}$, MDN with $K = 5$ Gaussian components, and policy logits $\pi_t^M \in \mathbb{R}^{B \times A}$ where A is the size of the legal-action set at step t (variable; the controller masks illegal entries to $-\infty$ before softmax). The JEPA target tensor matches z_{t+1} (no separate decoder); the controller value head outputs a scalar $V_t \in \mathbb{R}^B$. The checkpoint loader (`WorldModel.load`) is shape-tolerant: missing keys fall back to freshly-initialised weights, which lets us swap in checkpoints from earlier minor versions during ablation studies.

4.4 Active-inference LLM-fusion agent

The fused agent maintains a belief b_t over hidden state (opponent’s hand and library) using a particle filter constrained by the published decklist, and selects actions by minimising expected free energy:

$$\mathcal{F}(a) = \underbrace{-\mathbb{E}_{q(s'|a)}[R(s')]}_{\text{pragmatic}} - \underbrace{\text{H}[q(s'|a)] - \mathbb{E}[\text{H}[q(s'|s, a)]]}_{\text{epistemic (information gain)}} - \alpha \log p_{\text{LLM}}(a \mid \text{prompt}_t). \quad (12)$$

The LLM term provides a calibrated prior over the legal-action set; the controller of Section 4.3 provides R -estimates by short rollouts in the dream; and the KG provides hard combinatorial constraints (e.g., “no spell with mana cost > 3 in a deck whose curve peaks at 2”).

Worked example. Consider a mid-game decision with three legal actions: $a_1 = \text{cast } \textit{Counterspell}$ on a stack object, $a_2 = \text{pass priority}$, $a_3 = \text{activate a card-draw ability}$. With pragmatic-value horizon $H = 3$ controller rollouts the model returns $R(a_1) = +0.42$, $R(a_2) = -0.05$, $R(a_3) = +0.18$ (utilities normalised to $[-1, 1]$ over expected life-total swing). The particle filter has high posterior entropy on the opponent’s hand; the epistemic term $\text{H}[q(s'|a)] - \mathbb{E}[\text{H}[q(s'|s, a)]]$ peaks for a_3 (drawing reveals our top-of-library and lets us tighten the belief over our own future draws), giving an information-gain bonus of $+0.11$ versus ≤ 0.02 for the other actions. The LLM prior, asked “Given this board state, rank the three actions,” returns log-probabilities $(-0.6, -1.4, -1.0)$. With weights $(\alpha_R, \alpha_E, \alpha_{\text{LLM}}) = (1.0, 0.5, 0.3)$ the EFE scores are $\mathcal{F}(a_1) = -0.42 - 0.5 \cdot 0.02 - 0.3 \cdot (-0.6) = -0.25$, $\mathcal{F}(a_2) = +0.05 - 0.01 + 0.42 = +0.46$, $\mathcal{F}(a_3) = -0.18 - 0.055 + 0.30 = +0.065$. The agent picks a_1 (lowest free energy), matching the heuristic of countering rather than passing on a known threat. The same mechanism, with

epistemic weight α_E raised to 1.0, would have selected a_3 — the explore–exploit trade-off becomes a single hyperparameter.

4.5 Self-play training loop

We use a champion-vs-challenger promotion scheme:

Algorithm 1 Champion-vs-Challenger Self-Play (Phase A.5).

```

1: Initialise pool  $\mathcal{P}$  with  $K$  randomly-seeded agents.
2: Pick champion  $c^* \leftarrow \arg \max_{c \in \mathcal{P}} \text{ELO}(c)$ .
3: for round = 1, ...,  $R$  do
4:   Sample challenger  $c \sim \mathcal{P} \setminus \{c^*\}$ .
5:   Play  $N$  games  $c$  vs  $c^*$ , collect trajectories  $\tau$ .
6:   Update ELO for both; train all pool members on  $\tau$ .
7:    $w_r \leftarrow$  challenger win rate over the  $N$  games.
8:   if  $w_r \geq \theta$  then
9:     Promote:  $c^* \leftarrow c$ ; bump ELO by +50, demote former by −20.
10:  end if
11: end for

```

θ is the promotion threshold (default 0.55). The loop is implemented in `src/training/rl_trainer.py` and is exercised end-to-end by `scripts/train_pipeline.py`.

4.6 Agent zoo: algorithmic specification

The framework provides eight agents, all satisfying the same one-method contract `decide_action(state, legal_actions) → action` (see `src/agents/base_agent.py`). They span the full spectrum from purely combinatorial baselines to the V+M+C+JEPA+KG+LLM fusion of Section 4.3. We summarise each agent below; the engineering reference (exact class signatures, configuration knobs, fallback paths, and test pointers) is the companion technical report `docs/AGENTS_TECH_REPORT.md`.

Random (π^{rand}). A weighted sampler over legal actions with fixed type-class weights $w_{\text{land}} = 25$, $w_{\text{cast}} = 10$ (boosted to 30 for casts from the command zone), $w_{\text{atk}} = 8$, $w_{\text{pass}} = 1$. Used as a baseline floor; not deterministic per-instance because it shares the process-global RNG.

Heuristic (π^{heur}). Deterministic priority cascade

$$a^* = \arg \max_{a \in \mathcal{A}_{\text{legal}}} \text{prio}(a), \quad \text{prio} : \begin{cases} \text{PLAY_LAND} \mapsto 100 \\ \text{CAST_SPELL} \mapsto 90 \\ \text{DECLARE_ATTACKERS} \mapsto 80 \\ \text{DECLARE_BLOCKERS} \mapsto 70 \\ \text{ACTIVATE_ABILITY} \mapsto 60 \\ \text{PASS} \mapsto 1 \\ \text{CONCEDE} \mapsto 0, \end{cases}$$

with three tactical refinements: (i) counter-spell discipline filters `counter target` casts unless an opponent spell tops the stack, (ii) when `prefer_aggressive` is set the agent always picks an attack at the top tier, (iii) on the blocking step it chump-blocks the largest unblocked attacker with the smallest blocker. Tie-breaking uses a seeded `random.Random`, making it the strongest fully-reproducible baseline.

KG-Heuristic ($\pi^{\text{KG+heur}}$). Inherits π^{heur} but, when more than one `CAST_SPELL` is legal, re-ranks them through the knowledge graph. Let H be the agent’s hand and B its battlefield. For each candidate cast c ,

$$\text{score}_{\text{KG}}(c) = 10 \cdot \mathbf{1}[c \in \text{NearComboMissingPieces}(H \cup B)] + \sum_{p \in B} \text{strength}(\text{KG-edge}(c, p)),$$

where the indicator is computed by a single Cypher round-trip (`kg.detect_near_combos`) and synergy strength comes from the `HAS_SYNERGY` edges of ?? 1. The best-scoring cast replaces the cast subset in the legal-action list and the parent priority logic resumes, so `PLAY_LAND` can still beat `CAST_SPELL`. On any KG exception the agent latches `kg_disabled = True` and silently degrades to π^{heur} .

Ollama LLM (π^{LLM}). The LLM is consulted as a discrete posterior over the legal-action set, never as a free-form generator. Concretely, for each decision we serialise the board state s_t (life totals, hand sizes, lands, creatures) and an enumerated list $\{(i, a_i)\}_{i=0}^{|\mathcal{A}_{\text{legal}}|-1}$, prompt the model to “respond with ONLY the option number”, parse an integer $\hat{i}$ from the reply, and return $a_{\hat{i}}$. Formally,

$$\hat{i} = \pi_{\text{LLM}}(\text{prompt}_t), \quad a^* = a_{\text{clip}(\hat{i}, 0, |\mathcal{A}_{\text{legal}}|-1)}.$$

Illegal actions are unrepresentable by construction. We use Gemma 4-edge (`gemma4:e2b`) at temperature 0.2 via Ollama for a $\sim 1\text{--}4\text{s}$ budget per decision. On any HTTP or parse failure the agent falls back to π^{rand} and records the event in a per-game stats dictionary.

Active Inference (π^{AI}). Maintains a particle-filter belief b_t over the opponent’s hand and library top conditioned on the published decklist (`src/agents/opponent_model.py`). Action selection minimises the discrete-state expected free energy

$$\mathcal{F}(a) = -\mathbb{E}_{q(s'|a, b_t)}[R(s')] - \text{IG}(a, b_t), \quad \text{IG}(a, b_t) = H[q(s'|a, b_t)] - \mathbb{E}_{q(s|b_t)}[H[q(s'|s, a)]],$$

exposed as `ActiveInferenceModule.rank_actions(legal, state)`. A tactical override re-ranks `DECLARE_ATTACKERS` actions by target threat $\max(0, 40 - \text{life}) + 2 \cdot \text{commander_dmg_taken}$, and a counter-spell-aware filter drops non-mana `CAST_SPELL` actions when the opponent model places probability > 0.5 on a counter-spell being held.

World-Model agent (π^{WM}). Plays through the V+M+C+JEPA stack of Section 4.3. Every turn,

$$z_t = \text{Enc}_V(s_t), \quad g_t = \text{Enc}_{\text{KG}}(g, s_t), \quad (h_t, c_t) = \text{LSTM}_M([z_t; g_t], (h_{t-1}, c_{t-1})).$$

Two action-selection modes are available: a fast *direct policy* that calls the controller $C(z, h, A)$ and returns its argmax (or a temperature-scaled sample if `deterministic = False`), and a slower *dream search* that, for each candidate a_i , runs K rollouts of depth D in latent space using M , scores each rollout by the controller’s value head $V(z)$, and picks $\arg \max_i \frac{1}{K} \sum_k V(z_i^{(k, D)})$. Defaults are $K = 8, D = 10$. The hidden state (h, c) is persisted across turns so the dynamics model sees the full game-step trajectory.

LLM-Fusion agent (π^{fuse}). The strongest agent: a calibrated weighted sum of four scalar signals per legal action,

$$\text{score}(a) = \alpha_h s_{\text{heur}}(a) + \alpha_g s_{\text{KG}}(a) + \alpha_w s_{\text{WM}}(a) + \alpha_\ell \log p_{\text{LLM}}(a), \quad a^* = \arg \max_a \text{score}(a),$$

with default weights $(\alpha_h, \alpha_g, \alpha_w, \alpha_\ell) = (0.15, 0.15, 0.40, 0.30)$. The LLM signal is skipped when one action dominates the heuristic-plus-WM signal by more than `obvious_threshold = 0.8`,

which saves roughly 80% of LLM calls in practice. Setting any single weight to zero recovers the corresponding ablation; the canonical four-arm study (LLM-only, WM-only, KG-only, all four) is wired up by `scripts/run_matchups.py`.

Algorithm 2 LLM-Fusion decision (one priority window).

Require: state s_t , legal actions $\mathcal{A}$, world model (Enc_V, M, C) , KG $\mathcal{G}$, opponent model b_t , weights $(\alpha_h, \alpha_g, \alpha_w, \alpha_\ell)$

- 1: $\mathbf{s}_h \leftarrow \text{HeuristicScores}(s_t, \mathcal{A})$ $\triangleright \sim 50 \mu\text{s}$
- 2: $\mathbf{s}_g \leftarrow \text{KGScores}(s_t, \mathcal{A})$ $\triangleright$ combo + synergy bonuses
- 3: $z_t \leftarrow \text{Enc}_V(s_t)$; $\mathbf{s}_w \leftarrow \text{DreamSearch}(z_t, h_t, \mathcal{A}; K, D)$
- 4: **if** $\text{spread}(\mathbf{s}_h + \mathbf{s}_w) > \tau_{\text{obvious}}$ **then**
- 5: $\mathbf{s}_\ell \leftarrow \mathbf{0}$ $\triangleright$ skip the LLM call
- 6: **else**
- 7: $\mathbf{s}_\ell \leftarrow \log p_{\text{LLM}}(\cdot \mid \text{prompt}(s_t, \mathcal{A}))$
- 8: **end if**
- 9: **return** $a^* \leftarrow \mathcal{A}[\arg \max_i (\alpha_h \mathbf{s}_{h,i} + \alpha_g \mathbf{s}_{g,i} + \alpha_w \mathbf{s}_{w,i} + \alpha_\ell \mathbf{s}_{\ell,i})]$

Reasoning traces. Every agent writes a `ReasoningTrace` dict to `self.last_reasoning` before returning. The orchestrator flushes these to `runs/<out>/logs/game_NNNN.jsonl`, one line per decision, schema `{turn, phase, player_id, agent_kind, rationale, legal_action_count, chosen_index, top_candidates, scores, beliefs}`. This is the canonical telemetry format consumed by the post-hoc analysis notebooks and by the Streamlit visualisation panel discussed in Section 8.

5 Implementation

We keep implementation detail intentionally brief here because the scientific object of interest is the *interaction* between symbolic structure and learned planning, not the package layout of the repository. The released artifact instantiates four coupled layers: a rules-correct simulator, an ontology-backed card graph, a latent world model trained from self-play trajectories, and an agent layer that can combine heuristic, graph, world-model, and LLM signals. The code is modular enough that each layer can be ablated independently, but the paper’s main claim is about the resulting research design: MTG becomes a controlled setting for studying how symbolic priors, learned dynamics, and uncertainty-aware action selection cooperate.

Reproducibility. All randomness flows through a single `numpy.random.Generator` seeded by `config.seed`. Decklists are committed alongside the code under `examples/`. The pipeline (`scripts/train_pipeline.py`) has seven stages (infra check, KG build, GNN embedding, self-play, JEPA, dream evaluation, model evaluation) each individually re-runnable.

6 Preliminary Evaluation

We report preliminary results; a full benchmark suite (Section 8) is in development. Current evidence is of three kinds.

(i) Engine correctness. The repository contains ~ 30 pytest modules covering combat, the stack, triggered/static/replacement abilities, and end-to-end games. The whole-suite runtime is dominated by integration tests; unit tests under `tests/test_engine/` and `tests/test_knowledge/` pass on Python 3.11 and 3.12.

(ii) Knowledge-graph coverage. The OWL ontology v1.1 declares 17 top-level classes and 24 object/data properties. After Scryfall ingestion of a representative Standard environment ($\sim 5,000$ cards) the graph contains the cards, their oracle-text SBERT embeddings, and combo edges imported from Commander Spellbook.

(iii) Self-play promotion loop. The promotion loop of Algorithm 1 is exercised by two unit tests (`tests/test_rl_trainer.py`); both pass. End-to-end promotion behaviour — whether champions stabilise after $\mathcal{O}(10)$ rounds of $N = 20$ games — is the next planned experiment and is not reported here.

Checkpoint-benchmark status. We have *not yet* run a clean benchmark in this workspace showing archived, fully trained world-model checkpoints outperforming `heuristic`, `random`, or pure `llm` baselines. The current results establish that the post-fix world-model agent can play normally and tie the heuristic baseline on the tested Modern matchup, but the repository did not contain persisted training checkpoints at the time of writing. To close that gap we added a dedicated checkpoint benchmark path (`scripts/benchmark_trained_agents.py`) and an active-inference-inspired checkpoint selector, so the next training run can be evaluated as a true trained-vs-baseline comparison rather than a smoke test.

(iv) Pairwise ablation against the heuristic baseline. We use the in-tree `scripts/run_matchups.py` harness to play each candidate agent versus the strategy-aware `HeuristicAgent` on a fixed Modern matchup (Mono-Red Burn vs. Azorius Control), 16 games per cell with seat order swapped to remove going-first bias, `-max-turns 30`, seed 7. Results are summarised in Table 3. The KG-grounded heuristic and the LLM agent both clear 50 % against the baseline; the world-model agent ties; the LLM-WM-KG *fusion* agent currently underperforms its components, which we attribute to (a) its much larger per-decision latency biasing toward conservative passes when the cost-aware short-circuit fires, and (b) uncalibrated mixture weights $(\alpha, \beta, \gamma, \delta)$ in the σ scoring of §???. Calibration is queued in Section 8.

Agent vs. heuristic	Wins	Win-rate	Avg. turns	Avg. s/game
<code>kg_heuristic</code>	9 / 16	56.3%	6.4	0.32
<code>llm (Ollama)</code>	9 / 16	56.3%	5.9	1.02
<code>world_model</code>	8 / 16	50.0%	7.1	2.22
<code>llm_fusion</code>	5 / 16	31.3%	7.8	114.46

Table 3: Pairwise win-rate of each candidate agent versus the strategy-aware `HeuristicAgent` on a single Modern matchup (Mono-Red Burn vs. Azorius Control), 16 games per agent, seat-swapped, seed 7. Average wall time is per game on a single CPU; the LLM agent uses an Ollama-hosted Llama-3 model with `keep_alive` caching.

(v) Multi-agent round-robin (Standard, Swiss). We additionally ran a 7-agent Swiss tournament on the same Modern environment, 6 rounds, with Buchholz tiebreak (Table 4). The hand-tuned `HeuristicAgent` is currently undefeated; `ActiveInferenceAgent` and `LLMFusionAgent` share second place. The lower-bound `RandomAgent` sits at the expected level. The world-model and bare-LLM standings reflect the harness pre-fix — the snapshot of the corrected harness (1v1, Table 3) is the more reliable measurement, and a re-run of the multi-agent round-robin under the patched runner is queued in Section 8.

(vi) Commander pod. A 4-seat 8-round EDH pod (Krenko / Atraxa / Urza / Meren cores, seat rotations to remove turn-order bias) gave the standings in Table 5: the KG-grounded

Rank	Agent	W-D-L	Pts	Buchholz
1	heuristic	6-0-0	18	—
2	active_inference	4-0-2	12	45
3	llm_fusion	4-0-2	12	42
4	kg_heuristic	2-0-2	6	—
5	random	1-0-4	3	—
6	llm (pre-fix)	1-0-4	3	—
7	world_model (pre-fix)	0-0-4	0	—

Table 4: Standard 7-agent Swiss tournament, 6 rounds, seed 7. `llm` and `world_model` rows reflect the runner before the auto-concede fix described in §6 and are recorded here for transparency; their corrected pairwise numbers appear in Table 3.

heuristic was the clear winner, with `LLMFusionAgent` third despite its high per-turn cost. With only 4 distinct decks the pod is too small to draw strong conclusions, but it demonstrates that the system runs Commander games to completion at the expected $\sim 8\text{--}12$ turn length without crashing.

Rank	Agent	Wins	Pts
1	kg_heuristic	3	11
2	random	2	8
3	llm_fusion	1	5
4	heuristic	0	2

Table 5: Commander 4-seat pod, 8 rounds with seat rotations, seed 7. Sample size is small (8 games) and the deck pool is fixed at 4 cores — treat as a smoke test for end-to-end EDH play, not a strength claim.

(vii) **Debugging note: harness-induced auto-concede.** An earlier overnight run reported the LLM and world-model agents losing all 16 ablation games on turn 1. Investigation traced the cause to the rules engine always exposing `ActionType.CONCEDE` as a legal action (a debug affordance, see `rules_engine.get_legal_actions`). The Llama-3 prompt enumerated the legal-action indices but omitted `CONCEDE` from the rendered options, so a stray “1” in the model’s response collapsed onto the concede slot; the world-model controller, with effectively random initial weights, sampled the same slot stochastically. Filtering `CONCEDE` out of the candidate set inside both `LLMAgent.decide_action` and `WorldModelAgent.decide_action` restored normal play and produced the numbers in Table 3. We regard this as an instructive small example of how an under-specified action interface — not the policy itself — can dominate apparent agent quality, and have added a regression test to `tests/test_agents/`.

We deliberately do not claim “state of the art” numbers: we know of no comparable public baseline that plays full-Comprehensive-Rules MTG, and the contribution of this report is the system itself, plus the ontology and the released self-play infrastructure.

7 Discussion and Limitations

Scope. The current engine targets a Standard subset and an extended Commander subset; banding, phasing, planeswalker loyalty abilities, and exotic targeting restrictions remain partial. The ontology v1.1 covers core mechanics but does not yet encode every keyword from every set, and we do not currently extend it automatically per Standard rotation — a known limitation that motivates planned integration with the `OntologyExtender` tool (Section 8).

Compute. Defaults are tuned for a single-GPU workstation: $z \in \mathbb{R}^{256}$, the LSTM hidden size is 512, the JEPA predictor is a 4-layer pre-norm Transformer, and the LLM is `gemma4:e2b`. Larger pretrained encoders or stronger LLMs (e.g., `llama-3.1-70b`) would likely improve play strength but were not the focus of this report.

Active inference vs. RL. Active inference here is implemented primarily at the *decision* level (action selection conditional on world-model rollouts), not yet as the full learning objective for the world model itself. This is a deliberate simplification: the world model is still trained by the standard JEPA-style MSE+KL objective, which is much cheaper than full Bayesian model-evidence maximisation. The next step is to use active inference at *selection time* during training — treating checkpoint choice as an expected-free-energy problem that trades off win rate, robustness across opponents, and computational cost — before attempting a fully variational training objective.

LLM grounding. The LLM never produces game actions directly. It produces a posterior over the legal-action set provided by the engine; illegal actions are masked. This eliminates the dominant failure mode of LLM-only MTG bots (hallucinated cards, illegal plays) at the cost of a small expressivity loss.

8 Future Work

The following items are tracked in `IMPLEMENTATION_PLAN.md` and constitute the near-term roadmap:

1. **Strategy-aware and agent-driven mulligan policy.** The current keep/bottom heuristic is a single deterministic land-count rule shared by all agents. We are moving the decision through the agent interface (`MTGAgent.decide_mulligan`, `MTGAgent.bottom_cards`) so that aggressive, control, combo, and reactive strategies—and ultimately learned policies—can evaluate their own opening hands. JEPA-conditioned mulligan selection (rolling out a few imagined turns from each candidate hand) is the longer-term target.
2. **Priority-loop driven instants in the sync simulator.** The sync `GameSimulator` currently advances combat without exposing an instant-speed window to the non-active player; threading the existing `priority_loop` into combat steps will let counterspells and combat tricks fire mid-attack.
3. **Trained-agent benchmark.** Run the newly added checkpoint benchmark workflow on persisted world-model checkpoints so that we can answer the central empirical question directly: do trained world-model agents beat `random`, `heuristic`, `pure_llm`, and `active_inference` baselines on held-out matchups? The missing ingredient is no longer harnessing but archived checkpoints.
4. **Active-inference checkpoint selection.** Extend the current active-inference-inspired selector from post-hoc checkpoint ranking to an in-training promotion rule, where world-model candidates are selected not just by win rate but by an expected-free-energy objective combining performance, cross-baseline robustness, and decision cost.
5. **Ablation matrix.** Toggleable KG / dream / JEPA / opponent-model components with paired-game ELO comparison.
6. **Neural reasoner training.** A supervised + RL fine-tune of a graph-conditioned reasoner (`NeuralReasonerAgent`) replacing the SBERT card embedding with end-to-end GraphSAGE embeddings.

7. **Parallel self-play.** `asyncio.gather`-based parallel game execution to lift trajectory throughput, plus restoring `pytest-asyncio` so the async runner is exercised in CI.
8. **Belief-state visualisation.** JSON export and a Streamlit panel for the particle-filter posterior over the opponent’s hand.
9. **Commander politics.** Threat assessment across $N>2$ opponents, including a multi-agent extension of the active-inference objective, and the Commander variant of the mulligan (free first mulligan).
10. **Ontology evolution.** Wire the framework into the OntologyExtender tool ([Data Science Lab FHSWF, 2024](#)) so the OWL schema is updated semi-automatically per Standard rotation, with humans in the loop on debated axioms. At present the ontology is a static, hand-curated v1.1 artefact.

9 Conclusion

We have presented a research framework that approaches Magic: The Gathering not as “a harder Atari” but as a domain that fundamentally requires the combination of symbolic verification, structured knowledge, and learned imagination. The released codebase — engine, ontology, knowledge graph, V+M+C+JEPA world model, and active-inference LLM agent — is, to our knowledge, the first open system that puts these pieces together end-to-end on full Comprehensive-Rules MTG. We hope it is useful both as a benchmark for neuro-symbolic agents and as a sandbox for ideas in active inference, JEPA, and GraphRAG.

Reproducibility statement. All code, the OWL ontology, sample decklists, training scripts, and this report are available under MIT at <https://github.com/cuteredpwnda/opposition-agents-playing>

Acknowledgments

The authors thank the open-source maintainers of `open-mtg`, `Forge`, `XMage`, `Scrython`, `LangChain`, `LangGraph`, `Neo4j`, `n10s`, `APOC`, and `Ollama`, on whose work this project relies.

References

- D. Ha and J. Schmidhuber. World Models. *NeurIPS*, 2018.
- Y. LeCun. A Path Towards Autonomous Machine Intelligence. *OpenReview*, 2022.
- M. Assran et al. Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture. *CVPR*, 2023.
- D. Hafner, T. Lillicrap, J. Ba, and M. Norouzi. Dream to Control: Learning Behaviors by Latent Imagination. *ICLR*, 2020.
- D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap. Mastering Diverse Domains through World Models. *arXiv:2301.04104*, 2023.
- J. Schmidhuber. On Learning to Think: Algorithmic Information Theory for Novel Combinations of RL Controllers and Recurrent Neural World Models. *arXiv:1511.09249*, 2015.
- K. Irie et al. A Modern Self-Referential Weight Matrix that Learns to Modify Itself. *ICML*, 2022.
- K. Friston. The Free-Energy Principle: A Unified Brain Theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.

- N. Sajid, P. J. Ball, T. Parr, and K. J. Friston. Active Inference: Demystified and Compared. *Neural Computation*, 33(3):674–712, 2021.
- G. Tesauro. Temporal Difference Learning and TD-Gammon. *Communications of the ACM*, 38(3):58–68, 1995.
- D. Silver et al. Mastering the Game of Go without Human Knowledge. *Nature*, 550(7676):354–359, 2017.
- D. Silver et al. A General Reinforcement Learning Algorithm that Masters Chess, Shogi, and Go through Self-Play. *Science*, 362(6419):1140–1144, 2018.
- J. Schrittwieser et al. Mastering Atari, Go, Chess and Shogi by Planning with a Learned Model. *Nature*, 588(7839):604–609, 2020.
- O. Vinyals et al. Grandmaster Level in StarCraft II Using Multi-Agent Reinforcement Learning. *Nature*, 575(7782):350–354, 2019.
- M. Moravčík et al. DeepStack: Expert-Level Artificial Intelligence in Heads-up No-Limit Poker. *Science*, 356(6337):508–513, 2017.
- N. Brown and T. Sandholm. Superhuman AI for Heads-up No-Limit Poker: Libratus Beats Top Professionals. *Science*, 359(6374):418–424, 2018.
- P. I. Cowling et al. Ensemble Determinization in Monte Carlo Tree Search for the Imperfect Information Card Game Magic: The Gathering. *IEEE Transactions on Computational Intelligence and AI in Games*, 2012.
- W. L. Hamilton, R. Ying, and J. Leskovec. Inductive Representation Learning on Large Graphs. *NeurIPS*, 2017.
- D. Edge et al. From Local to Global: A Graph RAG Approach to Query-Focused Summarisation. *arXiv:2404.16130*, 2024.
- N. F. Noy and D. L. McGuinness. Ontology Development 101: A Guide to Creating Your First Ontology. Stanford Knowledge Systems Laboratory, 2001.
- H. Knublauch and D. Kontokostas (eds.). Shapes Constraint Language (SHACL). *W3C Recommendation*, 2017.
- Neo4j Labs. Neosemantics (n10s): RDF and Semantics Toolkit for Neo4j. <https://neo4j.com/labs/neosemantics/>.
- Data Science Lab, Fachhochschule Südwestfalen. OntologyExtender: A Multi-Agent Human-in-the-Loop Framework for Ontology Evolution. GitHub repository (work in progress), 2024. <https://github.com/DataScienceLabFHSWF/OntologyExtender>.
- R. Balestrieri, H. Van Assel, S. BuGhanem, and L. Maes. stable-pretraining-v1: Foundation Model Research Made Simple. *arXiv:2511.19484*, 2025. Software: <https://github.com/galilai-group/stable-pretraining>.
- L. Maes, Q. Le Lidec, D. Haramati, N. Massaudi, D. Scieur, Y. LeCun, and R. Balestrieri. stable-worldmodel-v1: Reproducible World Modeling Research and Evaluation. *arXiv:2602.08968*, 2026. Software: <https://github.com/galilai-group/stable-worldmodel>.
- L. Maes, Q. Le Lidec, D. Scieur, Y. LeCun, and R. Balestrieri. LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels. *arXiv preprint*, 2026. Software: <https://github.com/lucas-maes/le-wm>.